

Cell lines immunopeptidome analysis - Figure 3 - Supplementary file 1

Léa ROGUE

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Load libraries and functions

```
library(grid)
library(ggplot2)
library(reshape2)
library(UniProt.ws)
library(dplyr)
library(tidyr)
library(VennDiagram)
library(plotly)
library(ggrepel)
library(writexl)

map_genes_to_peptides <- function(df_summary) {
  df_sep <- df_summary %>%
    mutate(RowID = row_number()) %>%
    separate_rows(Prot_accessions, sep = ";\s*") %>%
    mutate(Prot_accessions = sub("-\\d+$", "", Prot_accessions))

  gene_mapping <- AnnotationDbi::select(up, keys = unique(df_sep$Prot_accessions),
    columns = "gene_primary", keytype = "UniProtKB")

  df_merged <- df_sep %>%
    left_join(gene_mapping, by = c(Prot_accessions = "Entry"))

  df_final <- df_merged %>%
    group_by(Seq) %>%
    summarise(Proteins = paste(unique(Prot_accessions), collapse = "; "),
      Genes = paste(unique(Gene.Names..primary.), collapse = "; "),
      .groups = "drop")

  df_summary %>%
    left_join(df_final, by = "Seq") %>%
    filter(Genes %in% l_CTA)
}

prep_df <- function(df) {

  df_summary <- df %>%
    group_by(Prot_accessions) %>%
    summarise(nb_sequences = n(), .groups = "drop")

  # Separate
  df_sep <- df_summary %>%
    mutate(RowID = row_number()) %>%
    separate_rows(Prot_accessions, sep = ";\s*") %>%
    mutate(Prot_accessions = sub("-\\d+$", "", Prot_accessions))

  # Mapping UniProt -> Gene(s)
  gene_mapping <- AnnotationDbi::select(up, keys = df_sep$Prot_accessions,
    columns = "gene_primary", keytype = "UniProtKB")

  # Merge and reconstruct
```

```

df_final <- df_sep %>%
  left_join(gene_mapping, by = c(Prot_accessions = "Entry")) %>%
  group_by(RowID, nb_sequences) %>%
  summarise(Proteins = paste(unique(Prot_accessions), collapse = "; "),
            Genes = paste(unique(Gene.Names.primary.), collapse = "; "),
            .groups = "drop") %>%
  arrange(desc(nb_sequences))

df_final <- df_final %>%
  group_by(Genes) %>%
  mutate(Genes_unique = if (n() > 1)
         paste0(Genes, "_", row_number()) else Genes) %>%
  ungroup() %>%
  mutate(Genes_unique = factor(Genes_unique, levels = Genes_unique)) %>%
  arrange(desc(nb_sequences))
return(df_final)
}

```

Read data

```

# Cell lines immunopeptidomic
df_ch_mhc1 <- read.table("../data/immunopep_CH2879.tsv", sep = "\t",
  header = T)
df_jj_mhc1 <- read.table("../data/immunopep_JJ012.tsv", sep = "\t",
  header = T)

# Filter on length
df_jj_mhc1 <- df_jj_mhc1[df_jj_mhc1$seq_len >= 8 & df_jj_mhc1$seq_len <=
  11, ]
df_ch_mhc1 <- df_ch_mhc1[df_ch_mhc1$seq_len >= 8 & df_ch_mhc1$seq_len <=
  11, ]

# CTA
l_CTA <- read.table("../data/CTA_list_clean.txt")$V1

# Read IEDB file
df_iedb_mhc1 <- read.csv("../data/epitope_table_export_human_MHC_I.tsv",
  header = T, sep = "\t")
df_pep_mhc1 <- df_iedb_mhc1[, c("Name", "Source.Molecule.IRI",
  "Molecule.Parent.IRI")]

# IEDB
pep_iedb_mhc1 <- unique(df_pep_mhc1$Name)
pep_iedb_mhc1 <- gsub(".*|\\|+.*", "", pep_iedb_mhc1)

# Read scatter data from Fig1E
df_scatter_hr <- read.table("../results/table_scatter_63_complete.tsv",
  sep = "\t", header = T)

# Read and prepare strong binders files
df_aff_jj <- read.table("../data/table_res_netmhc_all_JJ012.tsv",

```

```

sep = "\t", header = T)
df_aff_ch <- read.table("../data/table_res_netmhc_all_CH2879.tsv",
sep = "\t", header = T)

```

I. Venn diagram with CTAs

```

# CTA iedb
df_iedb_mhc1$ID <- gsub("http://www.uniprot.org/uniprot/", "",
df_iedb_mhc1$Molecule.Parent.IRI)
df_iedb_mhc1$ID <- sub("-.*", "", df_iedb_mhc1$ID)

# Transform ID in gene name
up <- UniProt.ws(taxId = 9606)
gene_mapping <- AnnotationDbi::select(up, keys = unique(df_iedb_mhc1$ID[df_iedb_mhc1$ID !=
""]), columns = "gene_primary", keytype = "UniProtKB")
colnames(gene_mapping) <- c("From", "ID", "Gene.Names..primary.")
df_iedb_mhc1_name <- merge(df_iedb_mhc1, gene_mapping, by.x = "ID",
all.x = TRUE)

# Select peptides from CTAs of IEDB
df_iedb_mhc1_cta <- df_iedb_mhc1_name[df_iedb_mhc1_name$Gene.Names..primary. %in%
l_CTA, ]

```

```

# Take CTAs from peptides in cell lines
df_final_pep_jj <- map_genes_to_peptides(df_jj_mhc1)
df_final_pep_ch <- map_genes_to_peptides(df_ch_mhc1)

# Clean pep iedb
pep_iedb_mhc1_cta <- unique(df_iedb_mhc1_cta$Name)
pep_iedb_mhc1_cta <- gsub(".*|\\+.*", "", pep_iedb_mhc1_cta)
pep_iedb_mhc1_cta <- unique(pep_iedb_mhc1_cta)

# Select new pep CH
new_pep_ch_mhc1 <- setdiff(df_final_pep_ch$Seq, pep_iedb_mhc1_cta)
grep_ch <- sapply(new_pep_ch_mhc1, function(x) any(grepl(x, pep_iedb_mhc1_cta)))
new_pep_ch_mhc1 <- new_pep_ch_mhc1[!new_pep_ch_mhc1 %in% names(grep_ch)[grep_ch]]
df_ch_new_pep_mhc1 <- df_ch_mhc1 %>%
filter(Seq %in% new_pep_ch_mhc1) %>%
distinct(Seq, .keep_all = TRUE)

# Select new pep JJ
new_pep_jj_mhc1 <- setdiff(df_final_pep_jj$Seq, pep_iedb_mhc1_cta)
grep_jj <- sapply(new_pep_jj_mhc1, function(x) any(grepl(x, pep_iedb_mhc1_cta)))
new_pep_jj_mhc1 <- new_pep_jj_mhc1[!new_pep_jj_mhc1 %in% names(grep_jj)[grep_jj]]

df_jj_new_pep_mhc1 <- df_jj_mhc1 %>%
filter(Seq %in% new_pep_jj_mhc1) %>%
distinct(Seq, .keep_all = TRUE)

# Clean IEDB pep

```

```

pep_iedb_extended_ch <- unique(c(pep_iedb_mhc1_cta, names(grep_ch)[grep_ch]))
pep_iedb_extended_jj <- unique(c(pep_iedb_mhc1_cta, names(grep_jj)[grep_jj]))

```

JJ

```

# Venn diagram
grid.newpage()
venn <- draw.pairwise.venn(area1 = length(pep_iedb_extended_jj),
  area2 = length(df_final_pep_jj$Seq), cross.area = length(df_final_pep_jj$Seq) -
    length(new_pep_jj_mhc1), category = c("IEDB", "JJ012"),
  fill = c("#2C7BB6", "#ff7f0e"), scaled = FALSE)
grid.draw(venn)

```

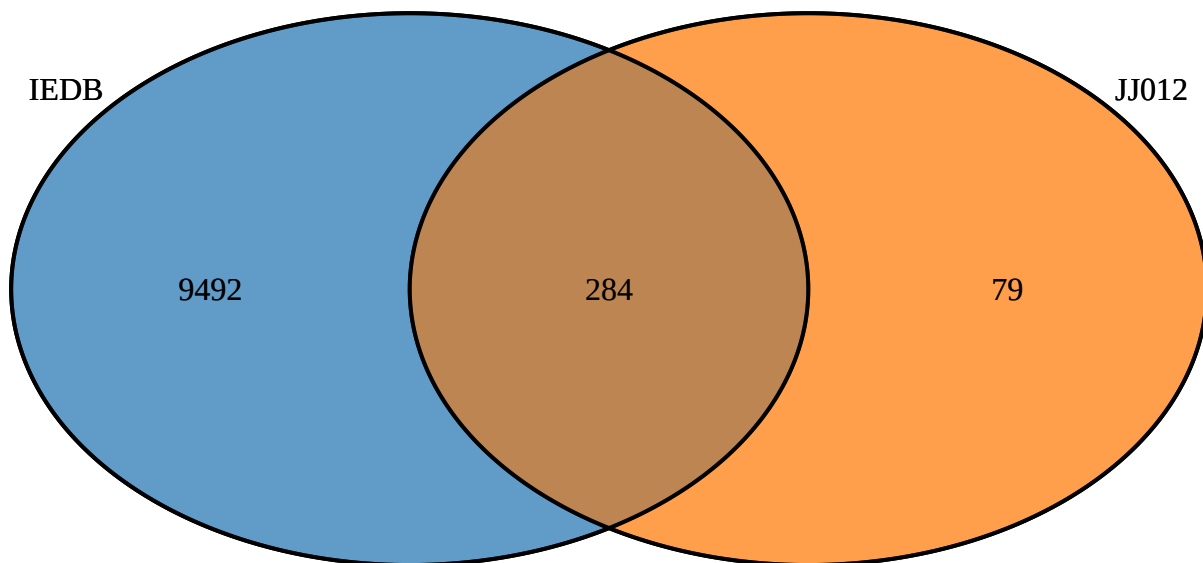


Figure 1: Venn plot JJ012 (MHCI)

CH

```
grid.newpage()
venn <- draw.pairwise.venn(area1 = length(pep_iedb_extended_ch),
  area2 = length(df_final_pep_ch$Seq), cross.area = length(df_final_pep_ch$Seq) -
    length(new_pep_ch_mhc1), category = c("IEDB", "CH2879"),
  fill = c("#2C7BB6", "#ff7f0e"), scaled = FALSE)
grid.draw(venn)
```

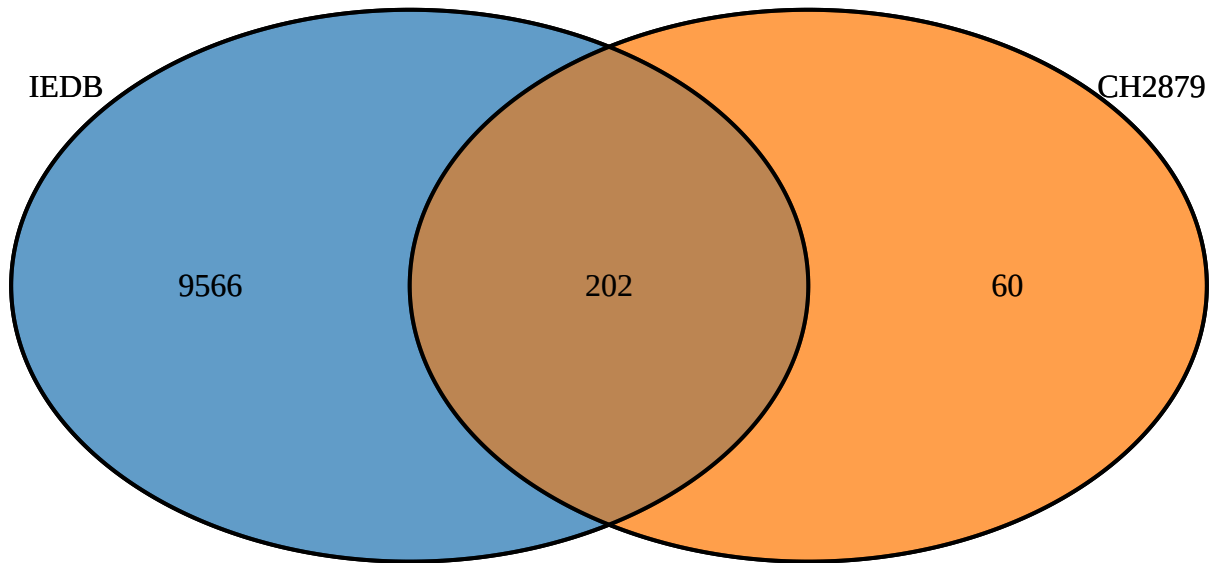


Figure 2: Venn plot CH2879 (MHCI)

Take CTA peptides for further analysis

```
# IEDB
df_iedb_cta <- df_iedb_mhc1_cta %>%
  mutate(Peptide = gsub(".*|\\|\\|+.*", "", Name)) %>%
  filter(Peptide %in% pep_iedb_extended_ch | Peptide %in% pep_iedb_extended_jj) %>%
  dplyr::select(Gene = Gene.Names..primary., Peptide) %>%
  distinct(Gene, Peptide)

# JJ012
df_jj_cta <- df_final_pep_jj %>%
  filter(Seq %in% df_final_pep_jj$Seq) %>%
  dplyr::select(Gene = Genes, Peptide = Seq) %>%
  distinct(Gene, Peptide)

# CH2879
df_ch_cta <- df_final_pep_ch %>%
  dplyr::select(Gene = Genes, Peptide = Seq) %>%
  distinct(Gene, Peptide)

# Merge
df_cta_all <- bind_rows(df_iedb_cta, df_jj_cta, df_ch_cta) %>%
  distinct(Gene, Peptide) %>%
  arrange(Gene, Peptide)

fasta_lines <- c(rbind(paste0(">", seq_len(nrow(df_cta_all))),
  df_cta_all$Peptide))
writeLines(fasta_lines, "../results/peptides_CTA_uniques_jj_ch_iedb.fa")
write.table(df_cta_all, "../results/peptides_CTA_uniques_jj_ch_iedb.tsv",
  row.names = FALSE, sep = "\t", quote = FALSE)
```

II. Histogram new peptides

CH2879

```
# Prepare data
df_ch_new_pep_mhc1_plot <- prep_df(df_ch_new_pep_mhc1)

# Plot
df <- df_ch_new_pep_mhc1_plot[df_ch_new_pep_mhc1_plot$Genes %in%
  l_CTA, ]
p <- ggplot(df, aes(x = Genes_unique, y = nb_sequences)) + geom_bar(stat = "identity",
  fill = "#ff7f0e") + theme_minimal() + labs(x = "Novel CTA proteins detected in CH2879",
  y = "Number of novel peptides") + theme(axis.text.x = element_text(angle = 90,
  hjust = 1, size = 5, color = "black"), panel.grid.major = element_line(color = "grey80",
  linewidth = 0.1), panel.grid.minor = element_line(color = "grey90",
  linewidth = 0.1)) + scale_x_discrete(labels = df$Genes)
p
```

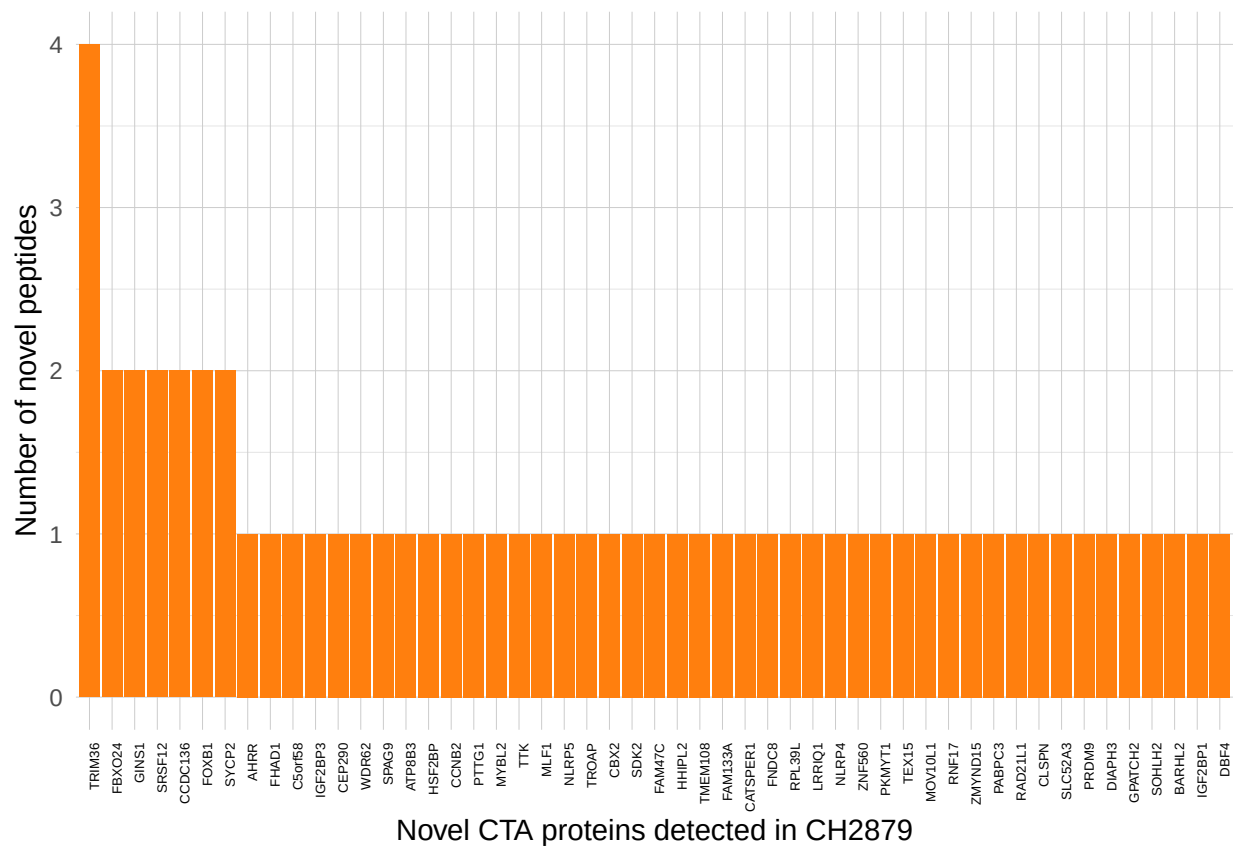


Figure 3: CTA new peptides histogram (CH2879, MHCI)

```
# ggsave('../results/histo_cta_ch.svg', plot = p, device =  
# 'svg', width = 4, height = 2)
```

JJ012

```
# Prepare data
df_jj_new_pep_mhc1_plot <- prep_df(df_jj_new_pep_mhc1)

# Plot
df <- df_jj_new_pep_mhc1_plot[df_jj_new_pep_mhc1_plot$Genes %in%
  l_CTA, ]
p <- ggplot(df, aes(x = Genes_unique, y = nb_sequences)) + geom_bar(stat = "identity",
  fill = "#ff7f0e") + theme_minimal() + labs(x = "Novel CTA proteins detected in JJ012",
  y = "Number of novel peptides") + theme(axis.text.x = element_text(angle = 90,
  hjust = 1, size = 5, color = "black"), panel.grid.major = element_line(color = "grey80",
  linewidth = 0.1), panel.grid.minor = element_line(color = "grey90",
  linewidth = 0.1)) + scale_x_discrete(labels = df$Genes)
p
```

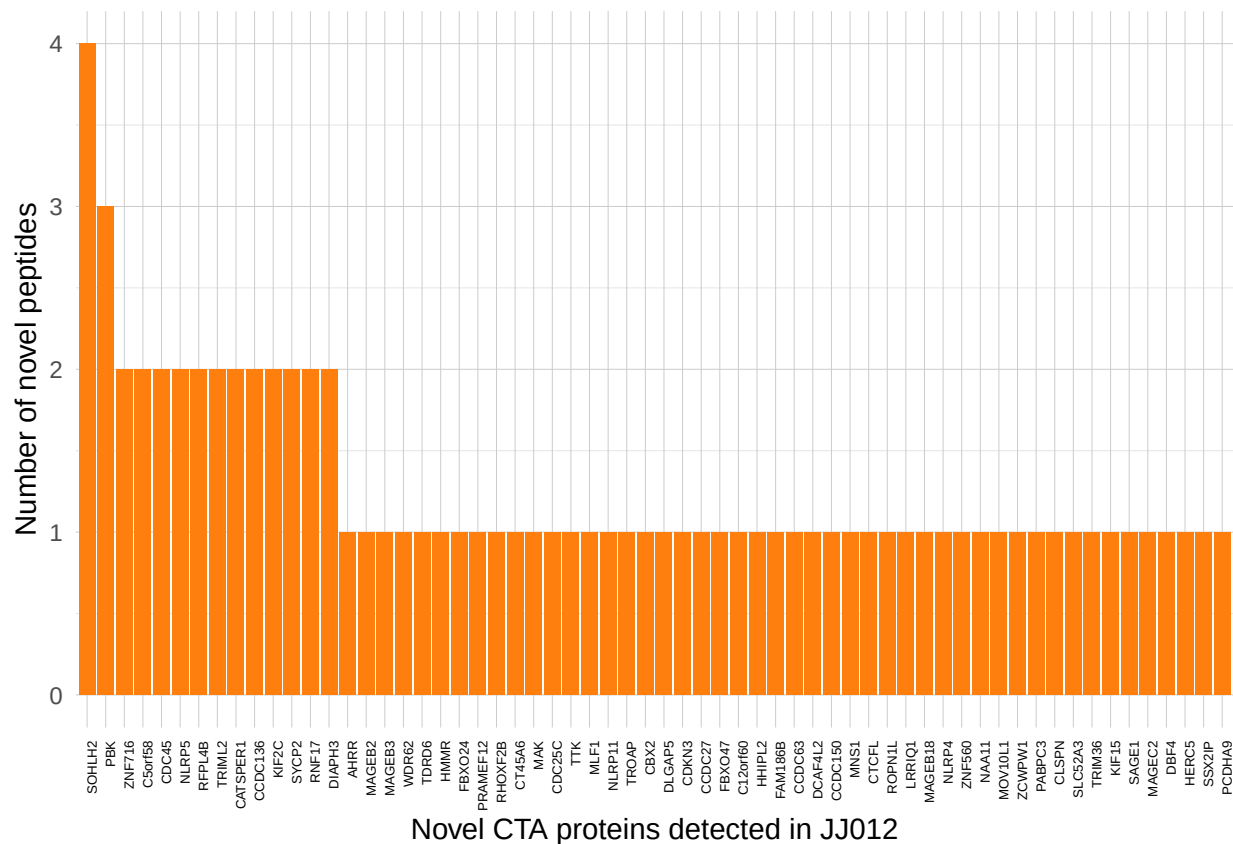


Figure 4: CTA new peptides histogram (JJ, MHCI)

```
# ggsave('../results/histo_cta_jj.svg', plot = p, device =
# 'svg', width = 4.5, height = 2)
```

III. Peptide ranks and scatter plots

```
# Select dominant HLA type
df_aff_ch_hlaa0101 <- df_aff_ch[df_aff_ch$Allele == "HLA-A0101" &
  df_aff_ch$Peptide %in% df_ch_mhc1$Seq, c("Peptide", "Affinity",
  "Rank")]

df_aff_jj_hlaa0201 <- df_aff_jj[df_aff_jj$Allele == "HLA-A0201" &
  df_aff_jj$Peptide %in% df_jj_mhc1$Seq, c("Peptide", "Affinity",
  "Rank")]
```

CH2879

Scatter plot from Fig1E

```
# Add known pep
pep_commons <- intersect(df_final_pep_ch$Seq, pep_iedb_mhc1_cta)

# Reunite known peptides
pep_commons2 <- names(grep_ch)[grep_ch]
pep_known_all <- union(pep_commons, pep_commons2)

# Dataframe with know CTA peptides and new peptides
df_iedb_new_pep_ch <- df_ch_mhc1 %>%
  filter(Seq %in% new_pep_ch_mhc1 | Seq %in% pep_known_all) %>%
  distinct(Seq, .keep_all = TRUE)

# Prepare df for scatter
df_iedb_new_pep_ch_anno <- prep_df(df_iedb_new_pep_ch)
df_merge <- merge(df_iedb_new_pep_ch, df_iedb_new_pep_ch_anno,
  by.x = "Prot_accessions", by.y = "Proteins")

# Add rank information
df_pep_ch_hlaa0101 <- df_merge[df_merge$Seq %in% df_aff_ch_hlaa0101$Peptide,
  ]
df_pep_ch_hlaa0101 <- merge(df_pep_ch_hlaa0101, df_aff_ch_hlaa0101,
  by.x = "Seq", by.y = "Peptide")
df_scatter_ch <- merge(df_scatter_hr, df_pep_ch_hlaa0101, by.x = "SYMBOL",
  by.y = "Genes")

# Plot
df_scatter_ch <- df_scatter_ch %>%
  mutate(Status = case_when(Seq %in% new_pep_ch_mhc1 ~ "New",
    Seq %in% pep_known_all ~ "Known"))

# Agregate per gene, if one is new => orange
df_scatter_ch_gene <- df_scatter_ch %>%
  group_by(SYMBOL, Mean_expression_tissues, Intensity, HR) %>%
  summarise(Status = ifelse(any(Status == "New"), "New", "Known"),
    Rank = min(Rank), .groups = "drop")
```

```

# Labeling genes
df_labels <- df_scatter_ch_gene %>%
  filter(Mean_expression_tissues < quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), Intensity > median(df_scatter_hr$Intensity)) %>%
  mutate(is_sb = Rank <= 0.5)

# For legends
df_legend <- data.frame(Mean_expression_tissues = c(NA, NA),
  Intensity = c(NA, NA), Status = c("Strong binders", "Others"))

# Plot
p <- ggplot(df_scatter_ch_gene, aes(x = Mean_expression_tissues,
  y = Intensity, color = Status, text = paste0("Gene: ", SYMBOL,
    "<br>Status: ", Status, "<br>Mean expression: ", round(Mean_expression_tissues,
    3), "<br>Intensity: ", round(Intensity, 3), "<br>HR: ",
    round(HR, 3), "<br>Rank: ", round(Rank, 3)))) + geom_point(size = 2,
  alpha = 0.8) + geom_point(data = df_legend, aes(x = Mean_expression_tissues,
  y = Intensity, color = Status), size = 0, alpha = 0, inherit.aes = FALSE) +
  scale_color_manual(values = c(Known = "#1f77b4", New = "#ff7f0e",
    `Strong binders` = "black", Others = "black"), labels = c(Known = "Known peptides",
    New = "Novel peptides", `Strong binders` = expression(bold("Strong binders (bold)")),
    Others = "Others"), breaks = c("Known", "New", "Strong binders",
    "Others")) + guides(color = guide_legend(override.aes = list(size = c(2,
  2, 0, 0), alpha = c(1, 1, 0, 0)))) + scale_x_log10(breaks = c(0.001,
  0.01, 0.1, 1, 10), labels = c("0.001", "0.01", "0.1", "1",
  "10")) + geom_hline(yintercept = median(df_scatter_hr$Intensity),
  linetype = "dashed", color = "black") + annotate("text",
  x = max(df_scatter_ch_gene$Mean_expression_tissues), y = median(df_scatter_hr$Intensity) -
    0.2, label = "Median\nexpression", size = 2.5) + geom_vline(xintercept = quantile(df_scatter_hr$
  0.75), linetype = "dashed", color = "black") + annotate("text",
  x = quantile(df_scatter_hr$Mean_expression_tissues, 0.75) +
    0.8, y = max(df_scatter_ch_gene$Intensity) + 0.2, label = "75th percentile",
  size = 2.5) + geom_text_repel(data = df_labels, aes(label = SYMBOL,
  fontface = ifelse(is_sb, "bold", "plain"), size = 3), color = "black",
  max.overlaps = 20, box.padding = 0.4, segment.color = "black",
  segment.size = 0.3, min.segment.length = 0) + scale_size_identity() +
  theme_minimal() + labs(x = "CTAs expression in normal tissues (Log10 TPM mean)",
  y = "CTAs expression in CHS (mean intensities)", color = "CH2879 HLA-A*01:01\npeptides")

```

p

```

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).

```

```

## Warning: ggrepel: 1 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps

```

```

# ggplotly(p, tooltip = 'text')
# ggsave('../results/scatter_plot_ch.svg', plot = p, width
# = 20, height = 15, units = 'cm')

```

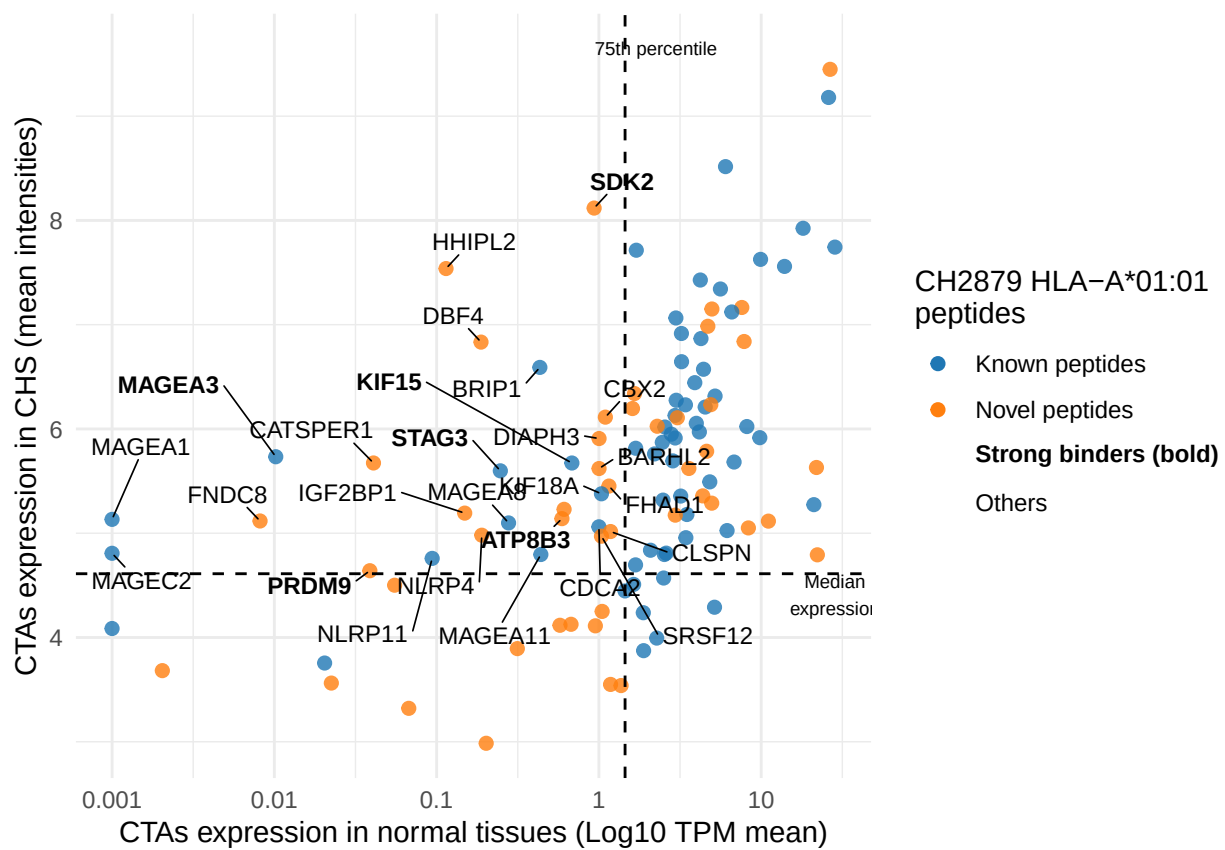


Figure 5: Scatter plot from Fig1E (CH2879)

Rank scatter plot

```
# Count number of peptides per rank
df_count <- df_scatter_ch %>%
  ungroup() %>%
  mutate(Status = as.character(Status)) %>%
  group_by(Rank) %>%
  summarise(nb_pep = n(), is_new = any(Status == "New"))

# Scatter plot
p <- ggplot(df_count, aes(x = nb_pep, y = Rank, color = is_new)) +
  geom_point(size = 2) + scale_color_manual(values = c(`FALSE` = "grey30",
  `TRUE` = "#ff7f0e"), labels = c(`FALSE` = "Known peptides",
  `TRUE` = "Novel peptides")) + geom_hline(yintercept = c(0.5,
  2), linetype = "dashed", color = "black") + annotate("text",
  x = max(df_count$nb_pep), y = 0.5 * 0.8, label = "Strong binders",
  hjust = 1, color = "black") + annotate("text", x = max(df_count$nb_pep),
  y = 2 * 0.8, label = "Weak binders", hjust = 1, color = "black") +
  labs(x = "CH2879 HLA-A*01:01 peptides number", y = "%Rank",
  color = "Category") + scale_y_log10(breaks = c(0.01,
  0.1, 1, 10, 100), labels = c("0.01", "0.1", "1", "10", "100")) +
  scale_x_continuous(breaks = seq(0, max(df_count$nb_pep),
  by = 2)) + theme_minimal()
```

p

```
# ggsave('../results/scatter_ch_a01.svg', plot = p)
```

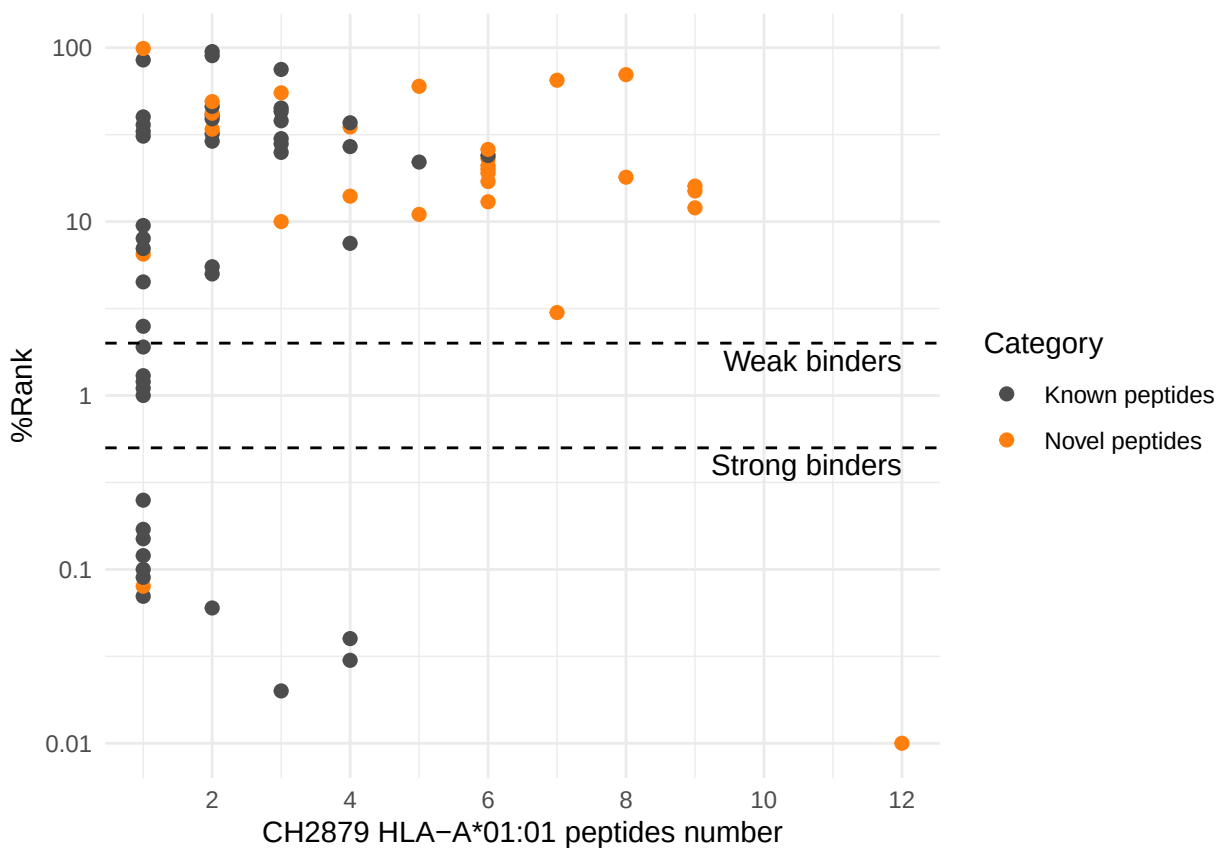



Figure 6: Rank scatter plot (CH2879)

JJ012

Scatter plot from Fig1E

```
# Add known pep
pep_commons <- intersect(df_final_pep_jj$Seq, pep_iedb_mhc1_cta)

# Reunite known peptides
pep_commons2 <- names(grep_jj)[grep_jj]
pep_known_all <- union(pep_commons, pep_commons2)

# Dataframe with know CTA peptides and new peptides
df_iedb_new_pep_jj <- df_jj_mhc1 %>%
  filter(Seq %in% new_pep_jj_mhc1 | Seq %in% pep_known_all) %>%
  distinct(Seq, .keep_all = TRUE)

# Prepare df for scatter
df_iedb_new_pep_jj_anno <- prep_df(df_iedb_new_pep_jj)
df_merge <- merge(df_iedb_new_pep_jj, df_iedb_new_pep_jj_anno,
  by.x = "Prot_accessions", by.y = "Proteins")

# Add rank information
df_pep_jj_hlaa0201 <- df_merge[df_merge$Seq %in% df_aff_jj_hlaa0201$Peptide,
]
df_pep_jj_hlaa0201 <- merge(df_pep_jj_hlaa0201, df_aff_jj_hlaa0201,
  by.x = "Seq", by.y = "Peptide")

# Prepare data for scatter
df_scatter_jj <- merge(df_scatter_hr, df_pep_jj_hlaa0201, by.x = "SYMBOL",
  by.y = "Genes")

# Plot
df_scatter_jj <- df_scatter_jj %>%
  mutate(Status = case_when(Seq %in% new_pep_jj_mhc1 ~ "New",
    Seq %in% pep_known_all ~ "Known"))

# Agregate per gene, if one is new => orange
df_scatter_jj_gene <- df_scatter_jj %>%
  group_by(SYMBOL, Mean_expression_tissues, Intensity, HR) %>%
  summarise(Status = ifelse(any(Status == "New"), "New", "Known"),
    Rank = min(Rank), .groups = "drop")

# Labeling genes
df_labels <- df_scatter_jj_gene %>%
  filter(Mean_expression_tissues < quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), Intensity > median(df_scatter_hr$Intensity)) %>%
  mutate(is_sb = Rank <= 0.5)

# Plot
p <- ggplot(df_scatter_jj_gene, aes(x = Mean_expression_tissues,
  y = Intensity, color = Status, text = paste0("Gene: ", SYMBOL,
    "<br>Status: ", Status, "<br>Mean expression: ", round(Mean_expression_tissues,
```

```

3), "<br>Intensity: ", round(Intensity, 3), "<br>HR: ",
round(HR, 3), "<br>Rank: ", round(Rank, 3))) + geom_point(size = 2,
alpha = 0.8) + geom_point(data = df_legend, aes(x = Mean_expression_tissues,
y = Intensity, color = Status), size = 0, alpha = 0, inherit.aes = FALSE) +
scale_color_manual(values = c(Known = "#1f77b4", New = "#ff7f0e",
`Strong binders` = "black", Others = "black"), labels = c(Known = "Known peptides",
New = "Novel peptides", `Strong binders` = expression(bold("Strong binders (bold)")),
Others = "Others"), breaks = c("Known", "New", "Strong binders",
"Others")) + guides(color = guide_legend(override.aes = list(size = c(2,
2, 0, 0), alpha = c(1, 1, 0, 0)))) + geom_hline(yintercept = median(df_scatter_hr$Intensity),
linetype = "dashed", color = "black") + annotate("text",
x = max(df_scatter_jj_gene$Mean_expression_tissues), y = median(df_scatter_hr$Intensity) -
0.2, label = "Median\nexpression", size = 2.5) + geom_vline(xintercept = quantile(df_scatter_hr$
0.75), linetype = "dashed", color = "black") + annotate("text",
x = quantile(df_scatter_hr$Mean_expression_tissues, 0.75) +
0.8, y = max(df_scatter_jj_gene$Intensity) + 0.2, label = "75th percentile",
size = 2.5) + geom_text_repel(data = df_labels, aes(label = SYMBOL,
fontface = ifelse(is_sb, "bold", "plain"), size = 3), color = "black",
max.overlaps = 20, box.padding = 0.4, segment.color = "black",
segment.size = 0.3, min.segment.length = 0) + scale_size_identity() +
scale_x_log10(breaks = c(0.001, 0.01, 0.1, 1, 10), labels = c("0.001",
"0.01", "0.1", "1", "10")) + theme_minimal() + labs(x = "CTAs expression in normal tissues (Log
y = "CTAs expression in CHS (mean intensities)", color = "JJ012 HLA-A*02:01\npeptides")

```

p

```

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).

```

```

## Warning: ggrepel: 11 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps

```

```

# ggsave('../results/scatter_plot_jj.svg', plot = p, width
# = 20, height = 15, units = 'cm')

```

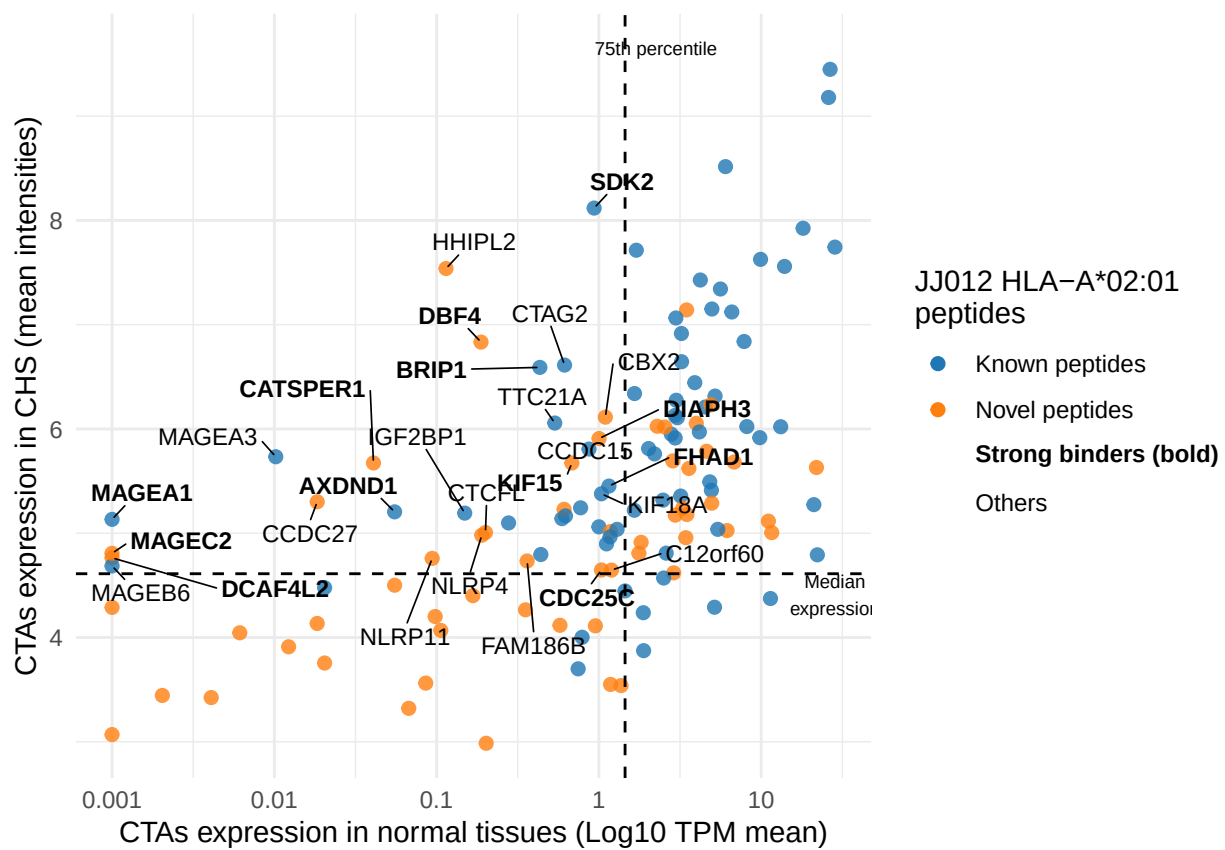


Figure 7: Scatter plot from Fig1E (JJ012)

Rank scatter plot

```
# Count number of peptides per rank
df_count <- df_scatter_jj %>%
  ungroup() %>%
  mutate(Status = as.character(Status)) %>%
  group_by(Rank) %>%
  summarise(nb_pep = n(), is_new = any(Status == "New"))

# Scatter plot
p <- ggplot(df_count, aes(x = nb_pep, y = Rank, color = is_new)) +
  geom_point(size = 2) + scale_color_manual(values = c(`FALSE` = "grey30",
`TRUE` = "#ff7f0e"), labels = c(`FALSE` = "Known peptides",
`TRUE` = "Novel peptides")) + geom_hline(yintercept = c(0.5,
2), linetype = "dashed", color = "black") + annotate("text",
x = max(df_count$nb_pep), y = 0.5 * 0.8, label = "Strong binders",
hjust = 1, color = "black") + annotate("text", x = max(df_count$nb_pep),
y = 2 * 0.8, label = "Weak binders", hjust = 1, color = "black") +
labs(x = "JJ012 HLA-A*02:01 peptides number", y = "%Rank",
color = "Category") + scale_y_log10(breaks = c(0.01,
0.1, 1, 10, 100), labels = c("0.01", "0.1", "1", "10", "100")) +
theme_minimal()
p
```

```
# ggsave('../results/scatter_jj_a02.svg', plot = p)
```

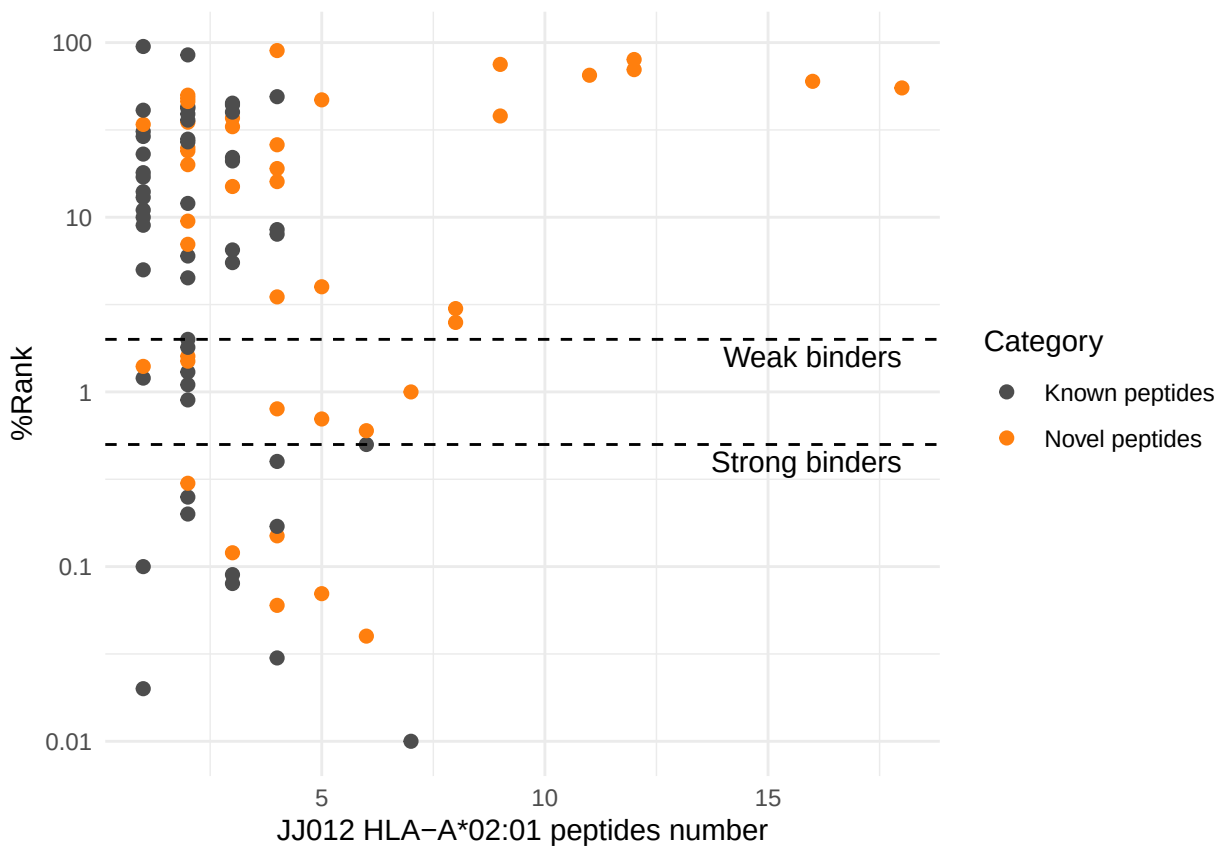


Figure 8: Rank scatter plot (JJ012)

IV. Supplementary data 1

```
# Select data
df_ch_supp <- df_scatter_ch[df_scatter_ch$Intensity > median(df_scatter_hr$Intensity) &
  df_scatter_ch$Mean_expression_tissues < quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), c("SYMBOL", "Intensity", "Mean_expression_tissues",
    "Cluster", "Seq", "Affinity", "Rank", "Status")]
df_jj_supp <- df_scatter_jj[df_scatter_jj$Intensity > median(df_scatter_hr$Intensity) &
  df_scatter_jj$Mean_expression_tissues < quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), c("SYMBOL", "Intensity", "Mean_expression_tissues",
    "Cluster", "Seq", "Affinity", "Rank", "Status")]

# Write in excel file
write_xlsx(list(JJ012_peptides_cadran = df_jj_supp, CH2879_peptides_cadran = df_ch_supp),
  path = "../results/fig3_supp.xlsx")
```